

Requirement 1: An employee (expert) cannot supervise or attend an auction they are already booked for

context Expert

inv NoDoubleBooking:

self.availabilities->forAll(a1, a2 |
a1 <> a2 implies a1.endTime <= a2.startTime or a2.endTime <= a1.startTime)

Requirement 2: An object must be either owned or up for auction, not both false

context ObjectOfInterest

inv MustBeOwnedOrAuctioned:

self.isOwned = true or self.isAuctioned = true

Requirement 3: Only one auction per location, time slot, and specialty

context Auction

inv NoAuctionOverlap:

Auction.allInstances()->forAll(a |
a <> self implies
a.location <> self.location or a.specialty <> self.specialty or
a.endTime <= self.startTime or self.endTime <= a.startTime)